MIL-P-12164E
29 March 1984
SUPERSEDING
MIL-P-12164D
23 October 1975

MILITARY SPECIFICATION

PLAYPIPE: NOZZLE, FIRE HOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a playpipe, to furnish a rigid extension to standard fire hose and to provide a gripping surface for firefighting personnel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-775	- Packaging of Hose, Hose Assemblies, Rubber, Plastic Fabric, or Metal (Including Tubing and Fittings, Nozzles, and Strainers).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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MIL-G-20241 - Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733 - Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28/10 - Screw Thread Standards for Federal Services, Section 10, American National Hose Coupling And Fire-Hose Coupling Threads.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

(Copies of specifications and standards required by contractor's in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/ASME - Boilers and pressure Vessels, Section IX, Welding Qualifications.
ANSI/AWS D1.1 - Structural Welding Code - Steel.

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM A 47 - Malleable Iron Castings.
ASTM B 16 - Free Cutting Brass Rod, Bar, and Shapes for use in Screw Machines.
ASTM B 26 - Aluminum-Alloy Sand Castings.
ASTM B 124 - Copper and Copper-Alloy Forging Rod, Bar, and Shapes.
ASTM B 584 - Copper Alloy Sand Castings for General Applications.
ASTM D 3951 - Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The playpipe shall be aluminum, brass, or malleable cast iron, as specified (see 6.2). The waterway bore of the playpipe shall be smooth and unobstructed and shall taper from 2-1/2 inches at the butt end to 1-1/2 inches at the tip end.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.3.1 Material deterioration and control. The playpipe shall be fabricated from compatible materials, inherently corrosion and derioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

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3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Malleable cast iron. Malleable iron castings shall conform to ASTM A 47, grade optional.

3.3.3 Cast aluminum. Aluminum castings shall conform to ASTM B 26, alloy B443.0 or D712.0.

3.3.4 Brass.

3.3.4.1 Brass bars. Brass bars shall conform to ASTM B 16.

3.3.4.2 Drop-forged brass. Drop-forged brass shall conform to ASTM B 124, copper alloy No. 377.

3.3.4.3 Cast brass. Brass castings shall conform to ASTM B 584, copper alloy No. 836, 852 or 857.

3.4 Threads. The butt end of the playpipe shall have a 2-1/2 inch x 7-1/2 NH internal threaded rigid connection, and the tip end shall have a 1-1/2 inch x 9 NH external threaded rigid connection. NH threads shall conform to FED-STD-H28/10.

3.5 Construction. The overall length of the playpipe shall be not less than 8 inches and not more than 11 inches. The playpipe shall be equipped with two diametrically opposed semiflexible handles swiveled around the playpipe at the tip and butt ends without binding (see figure 1 and 6.4). The swivel at the tip end shall have an integrally cast ladder hook. The handles shall be made of synthetic-covered rubber cable or reinforced dark-stained leather of two or more plies. The handles shall be reinforced at the swivel connection for rigid construction. The body of the playpipe shall be a one-piece casting. The butt end shall be equipped with rocker-type lugs. The lugs shall be designed for use with spanner wrenches.

3.6 Performance. The playpipe shall have a working pressure rating of 250 pounds per square inch gage (psig) and shall withstand a hydrostatic pressure of 500 psig without evidence of permanent deformation, cracks, or other evidence of damage or leakage.

3.7 Gaskets. One gasket shall be furnished to fit each internal-threaded connection. The gasket shall be commercial standard.

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3.8 Plating. All iron surfaces, including threaded and sliding surfaces, shall be chromium plated in accordance with the manufacturer's standard commercial practice.

3.9 Identification marking. The playpipe shall be identified in accordance with MIL-STD-130.

3.10 Treatment and painting. The portions of the playpipe normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard commercial practice.

3.11 Workmanship.

3.11.1 Castings and forgings. All parts, components, and assemblies of the playpipe which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.11.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.11.3 Welders and weldings.

3.11.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.
ANSI/ASME, Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with the ANSI/AWS or the ANSI/ASME code.

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3.11.3.2 Welding (aluminum). The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas shielded-arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or spatter.

3.11.3.3 Weldings. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the playpipe to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridgeing techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the playpipe shall be at the option of the contractor.

3.11.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article playpipe shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article playpipe shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Examination.

4.4.2.1 Individual. Each playpipe shall be examined for the critical defect specified in 4.5.1. Presence of a critical defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Test samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The playpipe shall be examined as specified herein for the following defects:

Critical

1. Holes in casting.

Major

101. Dimensions not as specified (see 3.1 and 3.5).
102. Materials not as specified (see 3.3).

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103. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
105. Contractor does not have documentation available for indentification of material, material finishes or treatments (see 3.3.1.2).
106. Internal diameter not uniform (waterway bore not smooth)(see 3.1).
107. Threads not as specified (see 3.4).
108. Components missing or not as specified (see 3.5).
109. Assembly not as specified (see 3.5).
110. Treatment and painting not as specified (see 3.10).
111. Workmanship not as specified (see 3.11).

Minor

201. Finish not as specified (see 3.8).
202. Swivel handles and swivel connections bind (see 3.5).
203. Gasket not as specified (see 3.7).
204. Identification marking missing, incomplete, or not legible (see 3.9).

4.5.2 Hydrostatic test. Connect a nozzle to the tip end of the playpipe. Establish a static pressure head of 500 psig at the inlet of the playpipe. Nonconformance to 3.6 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking for level A or level B shall be examined to determine compliance with the Quality Assurance Provisions of MIL-P-775. The commercial preservation, packing, and marking shall be examined to determine compliance with ASTM D 3951.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The playpipe shall be preserved in accordance with the level A preservation and packaging requirements of MIL-P-775.

5.1.2 Commercial. The playpipe shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

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5.2.1 Level A. The playpipe, preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-P-775.

5.2.2 Level B. The playpipe, preserved as specified in 5.1, shall be packed in accordance with the level B packing requirements of MIL-P-775.

5.2.3 Commercial. The playpipe, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with the requirements specified in MIL-P-775.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The playpipe is intended for use on hand-held 2-1/2 inch fire hose for directing a stream of water.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether aluminum, brass, or malleable cast iron playpipe is required (see 3.1).

c. When a first article is required for inspection and approval, and numbers of playpipes required (see 3.2).

d. When the welding process will not be at the option of the contractor (see 3.11.3.3).

e. Degree of preservation, and level of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article playpipe. The first article should consist of one or more playpipes. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

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6.4 Information figure. Figure 1 shows the type of a playpipe which is acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other playpipes which conform to this specification.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of the specification (see 3.3).

6.6 Classification relationship. The relationship between the classification of the playpipe included in this revision of the specification and the superseded document is as follows:

MIL-P-12164D	MIL-P-12164E
Type I	Discontinued
Type II	Identical, but without classification

Custodians:

Army - ME
Air Force - 99

Review activities:

Air Force - 84
DLA - CS

Preparing activity:

Army - ME

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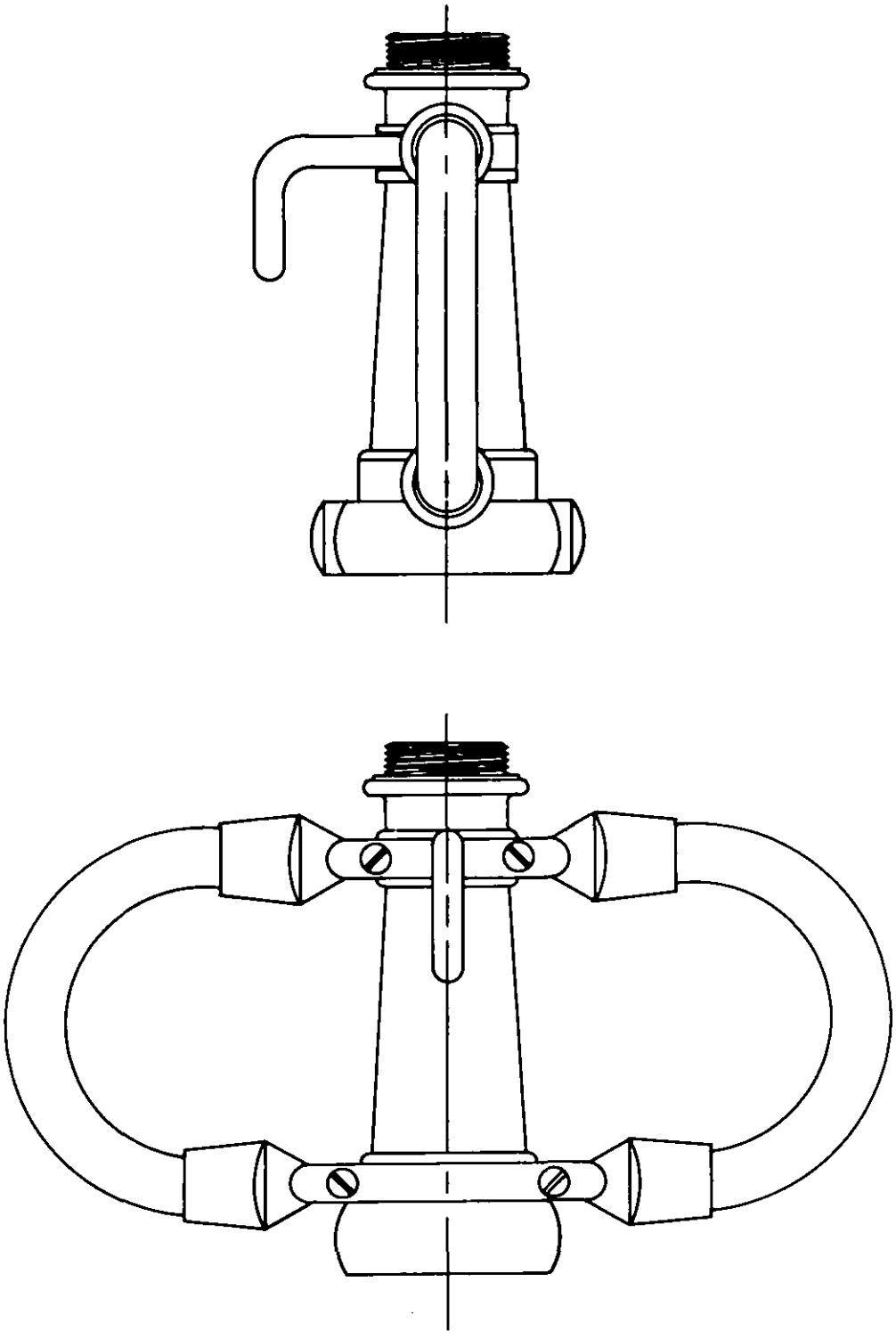


FIGURE 1. Playpipe, 2-1/2 inch (8-11 inch length).

X-1753C

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-12164E	Playpipe: Nozzle, Fire Hose

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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